

Session 2 · Deck 2.A3

The Stationary Distribution — and Where the Model Goes Next

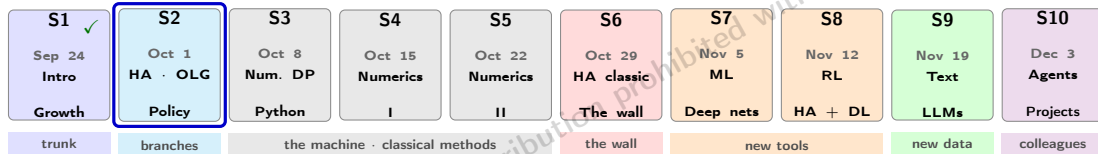
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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 2.A3



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 2.A1 why insurance fails · 2.A2 Aiyagari: production and the capital market · 2.A3 the stationary distribution, and where the model is taken next.

Two Innocent-Looking Transition Matrices

(a)

$$Q_a = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Stay where you are, forever. **Every** distribution is invariant — there is a continuum of them.

(b)

$$Q_b = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Flip every period. A unique invariant distribution exists, $(\frac{1}{2}, \frac{1}{2})$, but iterating from $(1, 0)$ **never converges** to it.

The point

Both are perfectly legitimate Markov chains. One breaks *uniqueness*, the other breaks *convergence*.

An Aiyagari solver would report an answer in both cases — a different answer depending on where you started.

In 2.A2 we wrote “let λ^* be the stationary distribution” as if that settled it. This deck asks what has to be true for that phrase to mean anything — and what your code is silently relying on.

The Object: a Markov Operator on Measures

The policy $a'(a, z)$ and the shock chain $\pi(z, z')$ induce a transition on $S = \mathcal{A} \times \mathcal{Z}$.

Acting on functions. For bounded measurable f ,

$$(Qf)(a, z) = \sum_{z'} \pi(z, z') f(a'(a, z), z').$$

Acting on measures (the adjoint Q^*). For Borel $\mathcal{B} \subseteq \mathcal{A}$,

$$(Q^*\lambda)(\mathcal{B}, z') = \sum_z \pi(z, z') \int_{\{a : a'(a, z) \in \mathcal{B}\}} d\lambda(a, z).$$

A **stationary distribution** is a fixed point: $\lambda^* = Q^*\lambda^*$.

Why this is not a contraction argument

In 1.B2 the Bellman operator was a contraction on *functions*, and Banach did the rest. Q^* acts on **probability measures** and is generally *not* a contraction — so we need different theorems.

Discretized, Q^* is a sparse matrix and λ^* its unit-eigenvalue eigenvector. A stochastic matrix always has eigenvalue 1 — so your code always returns *something*.

Three Properties, Three Guarantees

Result (Stokey, Lucas & Prescott 1989, Thm. 12.12)

If the Markov operator Q satisfies

1. the **Feller** property, then an invariant probability measure λ^* **exists**;
2. a **mixing** condition, then λ^* is **unique**;
3. **monotonicity**, then $Q^{*t}\lambda^0 \rightarrow \lambda^*$ for *any* initial λ^0 — **convergence**.

These are three *separate* guarantees. You can have existence without uniqueness (matrix (a)), and uniqueness without convergence (matrix (b)).

Each property is a statement about the **policy function** $a'(a, z)$, which is itself computed — so verifying them is part of verifying your solution.

In the Huggett economy of 2.A1 all three arrive bundled into one theorem. Huggett (1993), Thm. 2, via

Hopenhayn & Prescott (1987).

Feller $\Rightarrow$ Existence

The Feller property

Q maps continuous bounded functions to continuous bounded functions: $f \in C_b(S) \Rightarrow Qf \in C_b(S)$.

- In our setting this holds if the asset policy $a'(a, z)$ is **continuous** in a — which follows from concavity of u and the standard Berge/maximum-theorem argument of 1.B2.
- It also needs the state space to be effectively **compact**: when $\beta(1+r) < 1$ no household wants unbounded wealth, and $a \leq wz_{\max}/r$ suffices.
- Feller + compactness $\Rightarrow$ existence (Krylov–Bogolyubov).

Where it can fail

Discrete choices break continuity: occupational choice, a lumpy durable, a participation decision. The policy jumps, and Feller goes with it.

Such models can still have a stationary distribution — but it needs a separate argument.

Note the circularity: $\beta(1+r) < 1$ is needed for the bound, and $r < \rho$ is what the equilibrium delivers. The two arguments must be made together.

Mixing $\Rightarrow$ Uniqueness

“American Dream, American Nightmare”

From the poorest state $(\underline{a}, z_{\min})$ there is positive probability of reaching a higher state in finitely many periods — *and* from the richest state $(\bar{a}, z_{\max})$ there is positive probability of falling.

Formally — the *monotone mixing condition* — there are $s^*, \varepsilon > 0, N$ with $Q^N(\text{richest}, \{s \leq s^*\}) \geq \varepsilon$ and $Q^N(\text{poorest}, \{s \geq s^*\}) \geq \varepsilon$. Hopenhayn & Prescott (1987), Thm. 2.

- No region of the state space is permanently isolated, so the chain cannot support two separate invariant measures living on disjoint sets.
- Economically this is a statement about **mobility**: anyone can get richer, and anyone can get poorer.

Counterexample (a), revisited

$Q_a = I$: both states absorbing. Mixing fails, and every distribution is invariant.

The economic version: a policy function with a threshold a^* such that nobody below it ever crosses. The state space splits, and history determines the outcome.

This is exactly the **poverty trap** that quantitative models are sometimes built to produce. If you want one, you are deliberately breaking mixing — and uniqueness with it.

Monotonicity $\Rightarrow$ Convergence

Monotonicity

Q^* is order preserving: if λ first-order stochastically dominates λ' , then $Q^*\lambda$ dominates $Q^*\lambda'$.

- It holds when $a'(a, z)$ is **increasing** in both arguments and the shock chain is itself monotone (higher z today makes higher z' more likely, in the FOSD sense).
- Both are properties we can check on the computed policy — and both are properties the theory of 1.B2 gives us for the standard model.
- With monotonicity, iterating $\lambda^{n+1} = Q^*\lambda^n$ from any start converges. **This is the licence for simulation** — and in 2.A2 you saw it measured: five very different starts, terminal capital agreeing to 3.5×10^{-8} .

Counterexample (b), revisited

Q_b flips states every period. The unique invariant measure is $(\frac{1}{2}, \frac{1}{2})$, but the iterates cycle $(1, 0), (0, 1), (1, 0), \dots$ and never settle.

A non-monotone policy function can do the same thing in a real model.

Diagnostic you can actually run: plot $a'(a, z)$ and check it is increasing. If it is not, your stationary distribution may not be what your simulation converged to.

Back to the Crossing — and What Is Still Open

2.A2 named two assumptions behind the crossing and left them standing. Here they are, discharged. $K^s(r)$ must be **continuous** in r , and that needs:

- $a'(a, z; r)$ continuous in r — from the maximum theorem;
- λ_r continuous in r — from Stokey–Lucas–Prescott Thm. 12.13, which *presupposes* existence and uniqueness.

So the three properties are not decoration: they are what makes the equilibrium picture legitimate.

What we do know. Huggett (1997): with uninsurable labour-endowment risk, *any* steady-state capital stock lies above its deterministic counterpart. So $\beta(1 + r^*) < 1$ always.

What we do not know

Whether r^* is **unique** at the usual calibration.

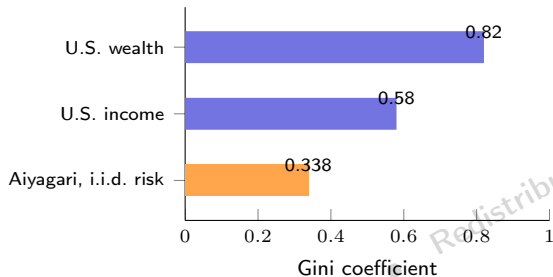
The sign of $\partial K^s / \partial r$ is ambiguous: a higher r raises the return to saving (substitution) but also makes a given target wealth easier to reach (income). Supply can bend backwards, and the curves can cross more than once.

Unique if the IES ≥ 1 (Light 2020; Achdou et al. 2022); not in general (Açıkgöz 2018).

Research in Practice

If your bisection on r finds a different root than someone else's, you report different policy conclusions from the same model.

How Much of the Data Does This Explain?



U.S.: SCF 2007, Díaz-Giménez, Glover & Ríos-Rull (2011). Model: the Stage-1A classical Aiyagari reference, Gini 0.338.

The honest verdict

The textbook model generates inequality from nothing but luck — and still reaches **well under half** the observed wealth Gini. It has essentially no top tail.

Why. Earnings risk alone is too small and too mean-reverting. Closing the gap requires an extra force: persistent or very high-variance earnings, heterogeneous returns, bequests, entrepreneurship, or preference heterogeneity. Each candidate is a **new state variable**.

Every Extension Adds a State

Question	Added state / choice	Why it matters
Who is uninsured, and what does that cost?	health status; insurance contract	medical spending is a large, persistent, uninsurable shock
Why is health insurance tied to the job?	job offer with or without coverage	insurance distorts who works and where
Who starts a business?	business ownership; entrepreneurial talent	the constraint binds on talent, not just on consumption
Why is wealth so concentrated?	heterogeneous returns; portfolio choice	return heterogeneity compounds, earnings risk does not
How does monetary policy work?	liquid vs. illiquid assets (HANK)	MPCs, not the interest-rate channel, do the work

The pattern from 1.B3, made concrete: the economics improves, the state space grows, and the computation gets harder in exactly the same step. That collision is what S6 calls **the wall**.

Research in Practice: Frictions on Top of the Engine

Research in Practice

The question: in the U.S. health insurance is tied to your job. What does that do to who works, who starts a business, and who saves?

- Health, health insurance and inequality (with Chen & Gu, *International Economic Review* 2025).
- Employment-based health insurance and misallocation (with Chivers & Villamil, *Review of Economic Dynamics* 2017).
- Uncertain medical expenses and aggregate labour supply (with Zhao, *JEBO* 2018).
- Tax subsidies to employer insurance and entrepreneurial talent (with Villamil, *Economic Theory* 2022).

Same engine, bigger state

Each of these is the Aiyagari household with extra states (health, job, business) and extra choices (hours, insurance, occupation).

And each one buys its realism with a dimension. Two continuous states are routine; five are a research project; the *distribution* as a state is 2.B1.

Takeaways

- The stationary distribution is a fixed point of a **Markov operator on measures**, not of a contraction on functions. It needs its own theory.
- **Feller** gives existence, **mixing** gives uniqueness, **monotonicity** gives convergence — and each fails in a two-state example you can write on a napkin.
- Those properties are what make the capital-market crossing of 2.A2 a legitimate argument, via continuity of $K^s(r)$.
- $r^* < \rho$ is known. **Uniqueness of r^*** is proved only for $\text{IES} \geq 1$; at $\gamma = 2$, report the bracket you searched.
- The model explains less than half the wealth Gini. Every serious fix adds a state variable — and the computational cost that comes with it.

Bridge: Now Let the Whole Economy Move

Everything in Block A rested on one assumption: **no aggregate shock**. That made λ^* constant and prices constant with it.

- So the distribution was a *fixed point* we solved for once, and then forgot.
- Households needed to forecast nothing: r and w were numbers.

Add a shock to total factor productivity and both statements fail. The distribution moves every period, prices move with it — and to choose today, a household must **forecast the entire distribution**.

After the break (Block B)

2.B1 Krusell–Smith: the distribution becomes a *state variable*, and something surprising happens — one number turns out to be almost enough.

2.B2 Finite lives and cohorts.

2.B3 The government as a player: Ramsey, time consistency, and policy as a fixed point in functions.